Quantum Dot-Based Direct X-Ray Detector for Mammography

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1. DESCRIPTION OF PURPOSE

X-Ray image sensors convert high energy photons into electric signals to visualize internal structures and are typically categorized as either indirect or direct conversion types. Indirect sensor employs a scintillator to convert X-Rays into visible light, which is then detected by a silicon-based image sensor. Although this technology is well established, it inherently suffers from reduced spatial resolution due to optical crosstalk, caused by the isotropic emission of light from scintillators. In contrast, direct sensor generates charge carriers directly from incident X-Ray photons, enabling high spatial resolution with minimal crosstalk.

The demand for higher spatial resolution in modalities such as mammography and dental imaging has driven the development of direct sensors using amorphous selenium (a-Se).[1] However, a-Se faces limitations due to its high ionization energy and low charge carrier mobility, which hinder its broad adoption.[2], [3] As a potential alternative, quantum dots (QDs) have recently gained attention. Their attenuation coefficient, work function, and electrical properties can be tuned by controlling material composition, particle size, and interdot spacing. Among various QD materials, PbS-based QDs are particularly attractive due to their strong absorption of X-Ray, owing to high atomic number and density of lead (Pb). Moreover, PbS QDs are widely used in solar cells and infrared sensors, therefore, this well-established method can be utilized.

In this study, we fabricated direct X-Ray detectors using PbS QDs and evaluated their performance within the mammography energy range. The device architecture was designed to optimize charge carrier collection efficiency with charge transporting layers. We also optimized the applied bias for dark current density and sensitivity, and achieving values of 1.95nA/cm^2 and 33μC Gy^{-1} cm^{-2}, respectively. These results indicate that the PbS QD-based detector delivers performance comparable to commercial a-Se devices.[1] Given its lower operating voltage and competitive metrics, the QD-based detector represents a promising solution for the mammography applications.

Keywords: PbS Quantum Dot, Direct X-Ray Detector, Mammography, Dark Current Density, X-Ray Sensitivity

2. METHODS

Absorption properties of PbS

Pbs QD-based detectors were designed, fabricated, and evaluated in the following steps. First, linear attenuation coefficients below 100keV were calculated using SpekPy for several X-Ray sensitive materials, including PbS, a-Se, CsI, HgI_2, and notable perovskites.[4] As shown in Figure 1(a), PbS and HgI_2 exhibit higher attenuation coefficients compared to the others within this energy range. In the mammography regime (photon energies below 28keV), PbS demonstrates particularly notable attenuation. In addition, to estimate the proportion of absorbed X-ray photons, absorbance as a function of thickness was derived from the attenuation coefficients. Figure 1(b) presents the absorbance of PbS, a-Se, HgI_2, and $CsPbBr_3$ within the mammography energy range. PbS shows a significantly higher absorbance, exceeding 90% at a thickness of 60μm. Considering practical density of PbS QDs, the QD layer thickness was set to 120μm.

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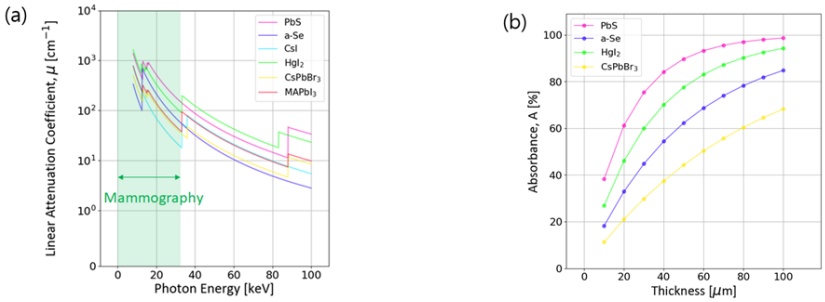

Figure 1. Absorption properties for several X-Ray sensitive materials. (a) Linear Attenuation Coefficient below 100keV, (b) Absorbance as function of thickness within the mammography energy region.

Sensor architecture

In contrast to the indirect conversion X-Ray detector – typically composed of amorphous silicon photodiodes on a thin film transistor (TFT) array – direct conversion detector requires direct conversion material on TFT array as illustrated in Figure 2. We used a conventional TFT array integrated with readout circuit. The pixel array consists of 256×256 pixels with a pixel pitch of 200µm. The digitization was performed with 16-bit resolution, with a system gain of 143e^-/DN and a full-well capacity of 940,000e^-. Based on the dark current density of PbS QDs, the frame rate was set to 50fps except during dark current measurements.

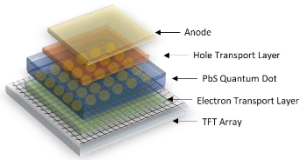

Figure 2. Schematic diagram of PbS quantum dot-based direct X-Ray detector.

The photoconductive layers were fabricated with solution-based monolithic integration of the transport and QD layers on the TFT array. The stack consists of thin film metal oxide for electron transport, a 120µm PbS QD layer, thin film hole transport layer, and a 80nm gold layer as top electrode. All transport layers are thin films that can have negligible X-Ray absorption.

Image Quality Evaluation

We evaluated dark current density and X-Ray sensitivity using image-based measurements. First, dark current density – which contributes to temporal noise and affects image quality – was estimated from images acquired at various frame times, following procedures described in previous literature.[5] Second, X-Ray sensitivity was evaluated using a tungsten-targeted X-Ray source with 50µm rhodium filter. Mammography-standard condition from IEC62220-1-2 Standard were followed:[6] Tube Voltage of 28keV, anode angle of 16 degree, 2mm of aluminum filter, source to image distance of 650mm, and tube current varied from 10mA to 60mA. The number of collected charges was calculated from the average pixel value, normalized by the dose measured with dosimeter. All measurements were repeated under various applied bias range from -5V to -25V, and optimal bias condition was determined.

3. RESULTS

Dark Current Density

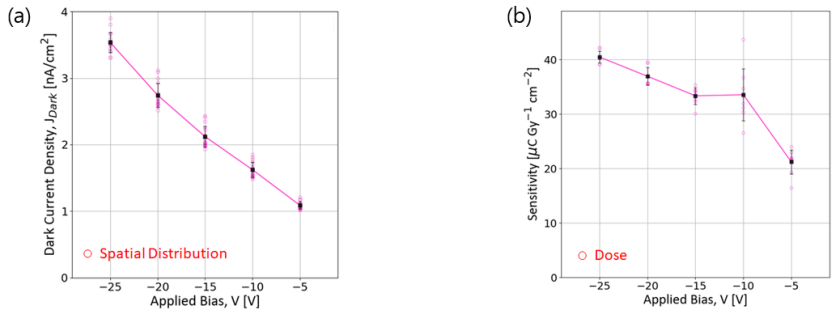

Figure 3. Characteristics of photoconductive layer analyzed through image as function of applied bias from -5V to -25V. (a) Dark current density, (b) X-Ray sensitivity

Figure 3(a) presents dark current density as a function of applied bias. Prior to calculating, each image was divided into 4×4 regions to identify spatial distribution of dark current density as indicated by the circles in the figures. Using the average pixel value and system gain, dark current density was obtained for each bias. The dark current densities are approximately 0.94nA/cm^2 at -5V, 1.95nA/cm^2 at -15V, and 3.38nA/cm^2 at -25V, respectively. As expected, the dark current density increases with increment of applied bias.

X-Ray Sensitivity

Figure 3(b) represents X-Ray sensitivity as a function of applied bias. Using the average pixel value, system gain, and dose, the X-Ray sensitivity was calculated. The sensitivity increases linearly from 21μC Gy^{-1} cm^{-2} to 40μC Gy^{-1} cm^{-2} with applied bias. While both dark current density and sensitivity show linear dependence on bias, sensitivity over temporal noise from dark current peaks at bias of -15V. Thus, the applied bias of -15V is optimal for maintaining image quality, especially under low dose conditions.

4. CONCLUSION

In this study, we developed and evaluated PbS QD-based direct X-Ray detectors for mammography applications. To determine the optimal operating conditions, we thoroughly analyzed the detector's performance under various applied biases. Our results show that both dark current density and X-Ray sensitivity increase with applied bias; however, an operating voltage of -15V offers the best image quality under low-dose conditions. At this bias, the dark current density and sensitivity were measured as 1.95nA/cm^2 and 33μC Gy^{-1} cm^{-2} respectively. These results are comparable to those of commercial a-Se detectors, while also benefiting from a lower operating bias. Thus, the PbS QD-based detector presents a promising alternative for mammography.

REFERENCES

[1] Kasap, Safa, et al. "Amorphous selenium and its alloys from early xeroradiography to high resolution X-ray image detectors and ultrasensitive imaging tubes." *Physica status solidi (b)* 246.8, 1794-1805 (2009); https://doi.org/10.1002/pssb.200982007

[2] Kasap, Safa O., and John A. Rowlands. "Direct-conversion flat-panel X-ray image sensors for digital radiography." *Proceedings of the IEEE* 90.4, 591-604 (2002); https://doi.org/10.1109/JPROC.2002.1002529

[3] Huang, Heyuan, and Shiva Abbaszadeh. "Recent developments of amorphous selenium-based X-ray detectors: a review." *IEEE Sensors Journal* 20.4, 1694-1704 (2019); https://doi.org/10.1109/JSEN.2019.2950319

[4] Poludniowski, Gavin, et al. "SpekPy v2. 0—a software toolkit for modeling x-ray tube spectra." *Medical Physics* 48.7, 3630-3637 (2021); https://doi.org/10.1002/mp.14945

[5] Weisfield, Richard L., and N. Robert Bennett. "Electronic noise analysis of a 127-um pixel TFT/photodiode array." *Medical Imaging 2001: Physics of Medical Imaging*. Vol. 4320. SPIE (2001); https://doi.org/10.1117/12.430955

[6] IEC 62220-1-2: Medical Electrical Equipment – Characteristics of Digital X-Ray Imaging Devices – Part 1-2: Determination of the Detective Quantum Efficiency – Detectors Used in Mammography.